

IPv6 Addressing

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Disclaimer

Sources and Links

- networkacademy.io - IPv6

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Introduction to IPv6

RFC1883

- larger 128 bit (32 bit hexadecimal) address space, allowing multi level subnetting and more efficient allocation from the regional internet providers
- end to end security, IPsec is built in, IPv6 has a header extension, that ease the implementation of encryption and authentication
- QoS
- stateless and statefull host addressing (SLAAC) - hosts can be addresses himself and using the network, if the dhcp failed
- simpler, more efficient header format >> non essential parts compared to IPv4 was removed, making it more efficient for intermediate routers
- extensibility: allow adding extensions headers after IPv6 header
- ARP and Broadcast replaced by ICMPv4 Neighbor Discovery, which use multicast messages instead of broadcast
- multiple IPv6 addresses per device on the same subnet, which allow better security, greater privacy, creates possibility for additional network features

Important, that all network features such as protocols and functions, that based on the IPv4 addressing, its not work under IPv6 due of the length of the addresses. Therefore, to migrate IPv6 in all layer, lead to rewrite all involved protocols and functions.

Representing IPv6 Addresses

- abbreviation

IPv4 Header vs IPv6 Header

IPv6 Header

Version	Traffic Class	Flow Label	
Payload Length		Next Header	Hop Limit
Source Address			
Destination Address			

IPv4 Header

Version	IHL	Type of Service	Total Length	
Identification			Flags	Fragment Offset
TTL	Protocol	Header Checksum		
Source Address				
Destination Address				
Options			Padding	

Legend

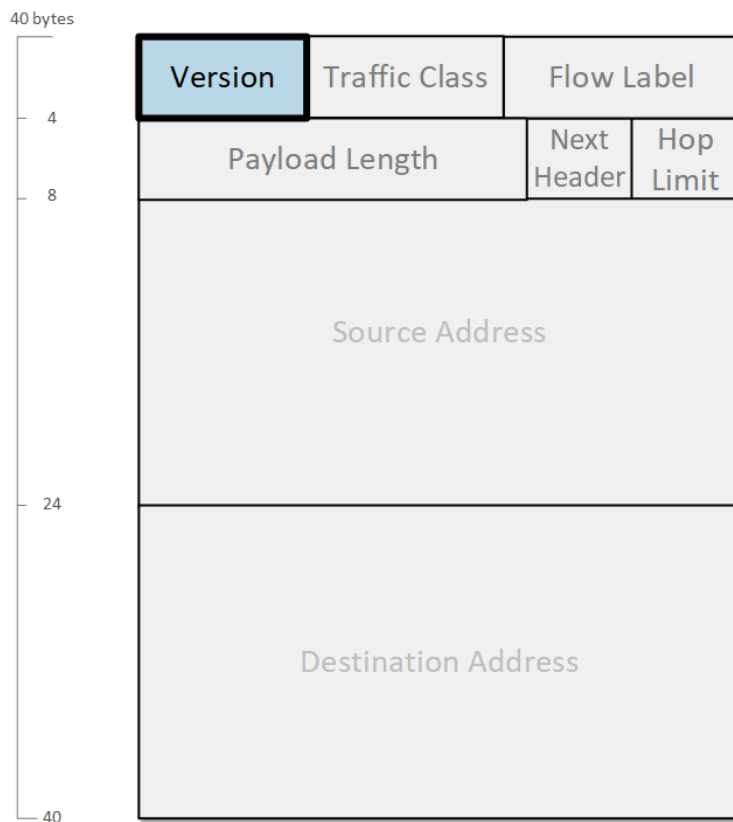
	Fields kept in IPv6
	Fields kept in IPv6, but name and position changed
	Fields not kept in IPv6
	Fields that are new in IPv6

IPv6 Header Format

- divided into two main Header, Main and Extensions Headers
- fewer fields make more efficient and faster to process it
- fixed 40 bit size

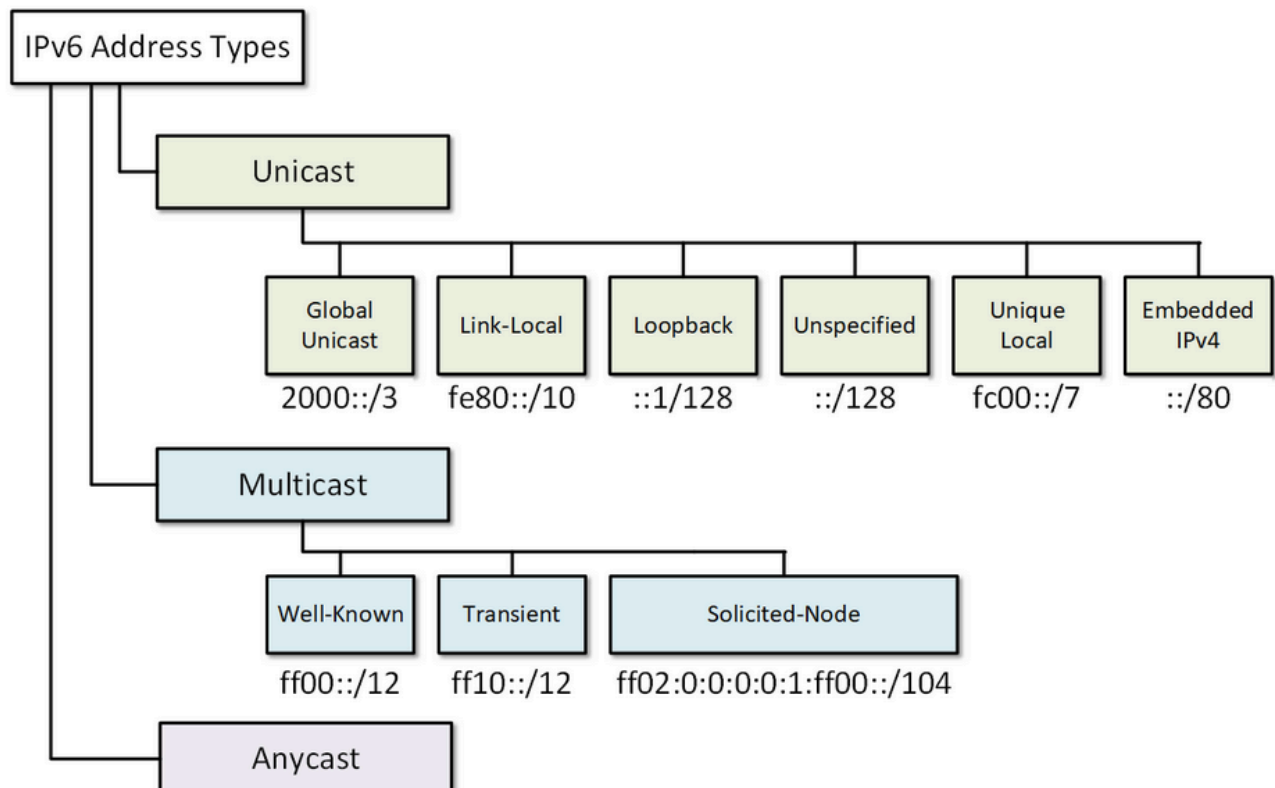
Version Field

- in IPv4 is set to 4, sure by IPv6 its 6



IPv6 Address Types

Overview of IPv6 Types



Unicast Addresses

- general type of addresses, which are unique identifier for a single network interface
- allow one- to one communication between devices
- down under, we discuss the different types of unicast addresses

Multicast Addresses

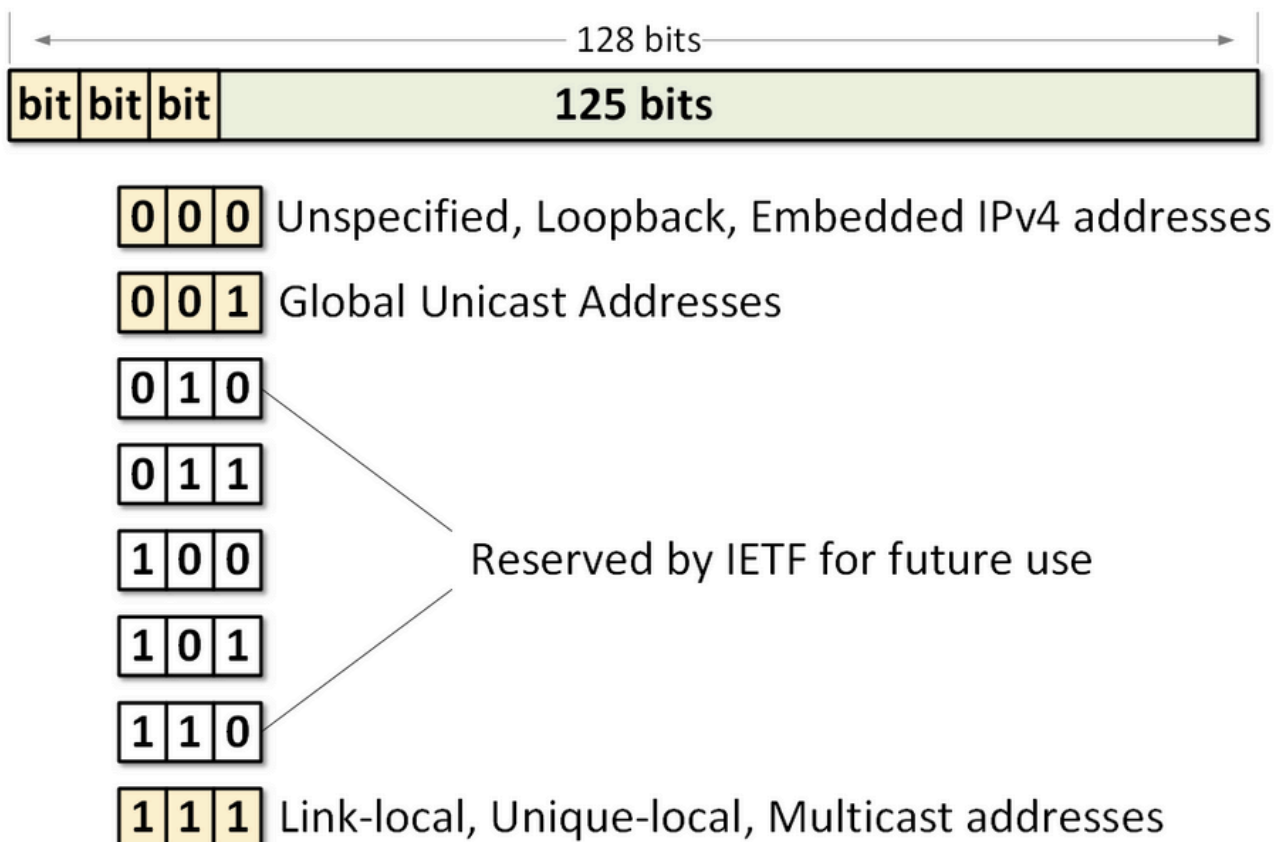
- network identifier for a set of interfaces, belonging to a different IPv6 enabled nodes
- packet delivered for all interfaces, that are identified by that address
- one to many

Anycast Addresses

- network layer identifier, for a set of interfaces, belonging to different IPv6 enabled nodes
- but unlike in multicast, the packets are delivered only for the closest destination in term of a routing metric

****No broadcast type in IPv6, since is replaced by using multicast addresses!!**

IPv6 Address Spaces



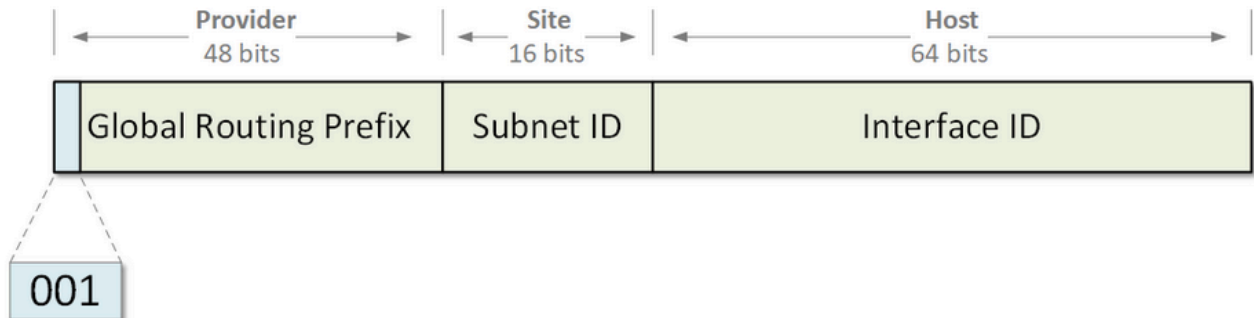
Group/Category	Prefix/Range (in CIDR notation)	Usage
Global Unicast	2000::/3 to 3FFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF	Public Address This is the largest block of IPv6 internetwork addresses assigned by IANA to ISPs and other large organizations.
Link-Local Unicast	FE80::/10 /64 is most common	Single-Interface Communication Used for communication between devices on the same link. Used commonly on point-to-point and multiaccess networks. Devices use link-local addresses for neighbor discovery and automatic configuration (SLAAC). Routers use link-local addresses for routing protocol messages and for local delivery of packets.
Unique Local Unicast (ULA)	FC00::/7 /8 and FD00::/8	Private Address Analogous to IPv4's private address ranges (10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16) for use within an organization. Designed for use within a site or organization. Not routable on the public Internet.
Multicast	FF00::/8	One-to-Many Communication Used for communication between a single source and multiple destinations.

Group/Category	Prefix/Range (in CIDR notation)	Usage
		multiple destinations simultaneously. Includes known and unknown. All Network. All Resources.
Loopback	::1/128	The e 127.0 Used send itself t local s access
Unspecified	:::/128	Represent address by a h initiali a rout the at route.
Embedded/Transition	2001::/32 (Teredo), 2002::/16 (6to4), ::/96 (IPv4-compatible)	Used trans mech tunne over I (e.g., Teredo these mech now c disco
Reserved/Special	0000::/8 (Excluding Loopback/Unspecified) and FFFF::/8	Curre for fut speci functi

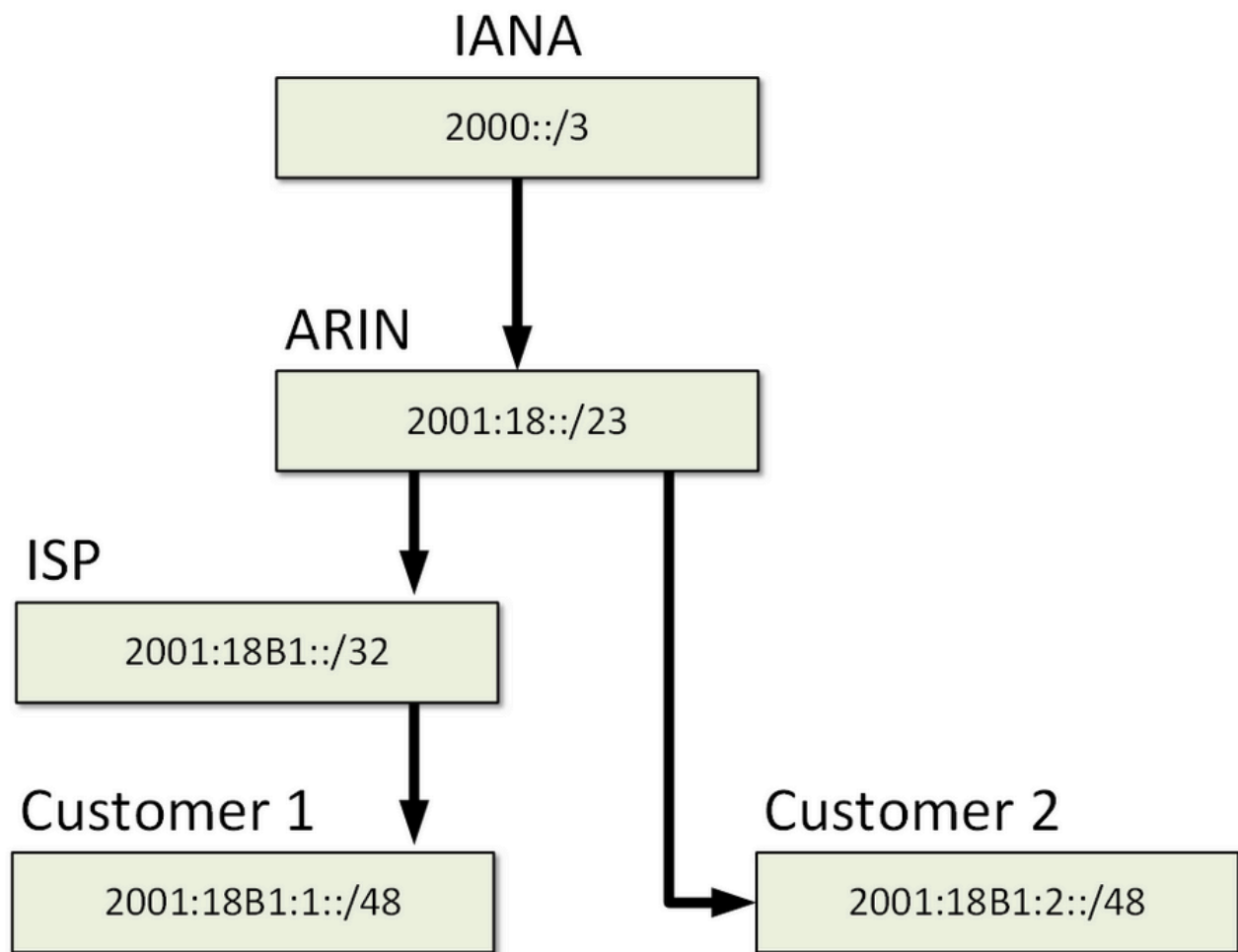
Unicast Addresses

Global Unicast Addresses - Public IPv4

- same as the public IPv4 Addresses, they are used in the internet also public routable addresses
- unique

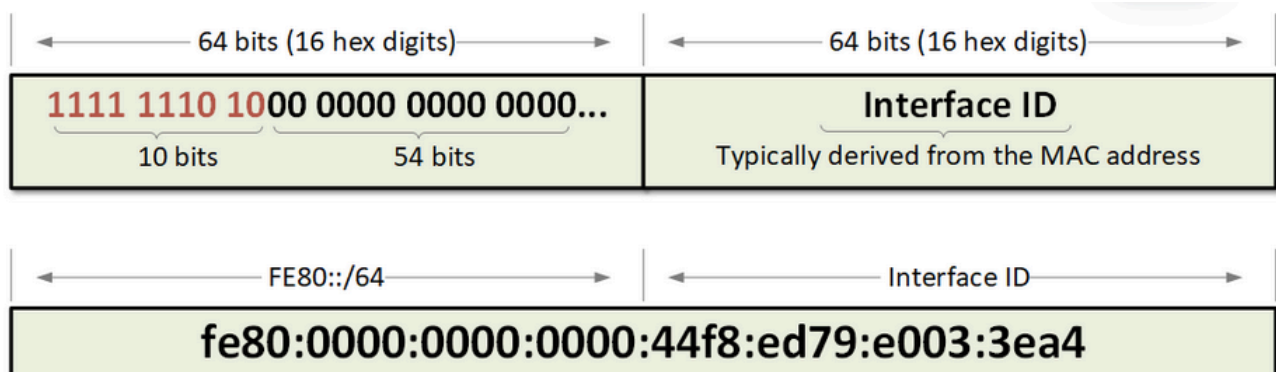


- aggregatable global unicast addresses are part of the global routing prefix
- structure of these addresses are enable for aggregation of routing entries to achieve a smaller global IPv6 routing table
- all global unicast address start with binary value 001 /3
 - 48 bit global routing prefix
 - 16 subnet ID also refered Site-Level Aggregator (SLA)
 - 64 bit interface ID



Link Local Addresses - APIPA

- APIPA - local, not routable, auto-configured
- unique for each LAN
- used for temporary LAN, to still have file and resource sharing
- link local prefix **FE80::/10** (first 10 bits equal to 1111 1110 10)



Loopback

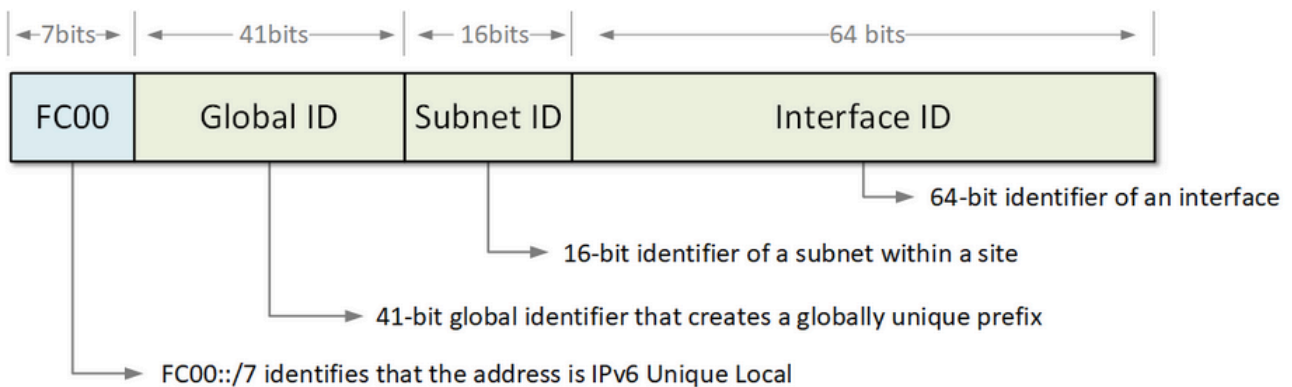
- same as by IPv4, no physical representation
- always up and running

- packages are returned "looped" to the source interface to testing the TCP/IP networking stack
- **0:0:0:0:0:0:0:1/128 is reserved for loopback identifier**
- **shortened to ::1/128**

Unspecified

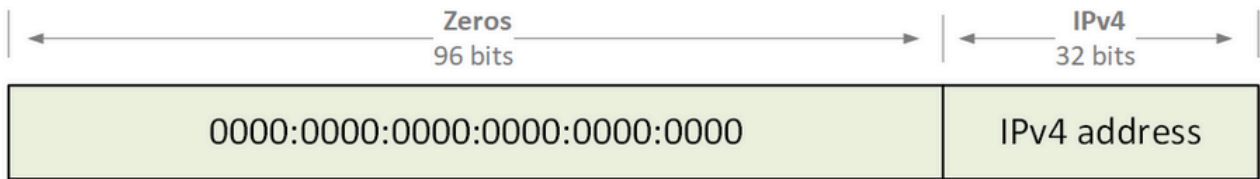
- in both IP protocol stack, the address with all binary to set 0
- IPv4 its 0.0.0.0/32
- IPv6 its 0:0:0:0:0:0:0:0 or shortened as ::/128
- routers do not forward packets with a source or destination address set to unspecified address.

Unique Local Addresses - Private IP Range



- same as IPv4 private addresses, its used on local network and routable in local network only
- globally unique prefix, will no conflict with other IPv6 global prefixes, no overlapping
- threatred as regular global IPv6 addresses
- Internet Service Provider independent address space
- nearly globally unique, designed to replace site-local addresses
- allow routing on multiple local networks

Embedded IPv4 and IPv6



192.168.45.1 embedded in IPv6 = ::C0A8:2D01

- unicast address which **has only zeros in the first 96 bit**
- the rightmost **32 bit contain the IPv4 address**
- IPv4 embedded in hex digits like A.B.C.D >> A:B:C:D
- its used in automatic tunnels, supporting both IP protocol stack
-

Multicast Addresses

Multicast - FF

- Technic used to send one to many
- destination is a set of interfaces, identified by one single multicast address, know as multicast group
- one to many addresses, **starting always with FF, prefix ff00::/8** (which is equivalent to the IPv4 multicast address space of 224.0.0.0/4)
- therefore the **leftmost bits are all set of 11111111**
- **There aren't broadcast addresses in IPv6.**
-

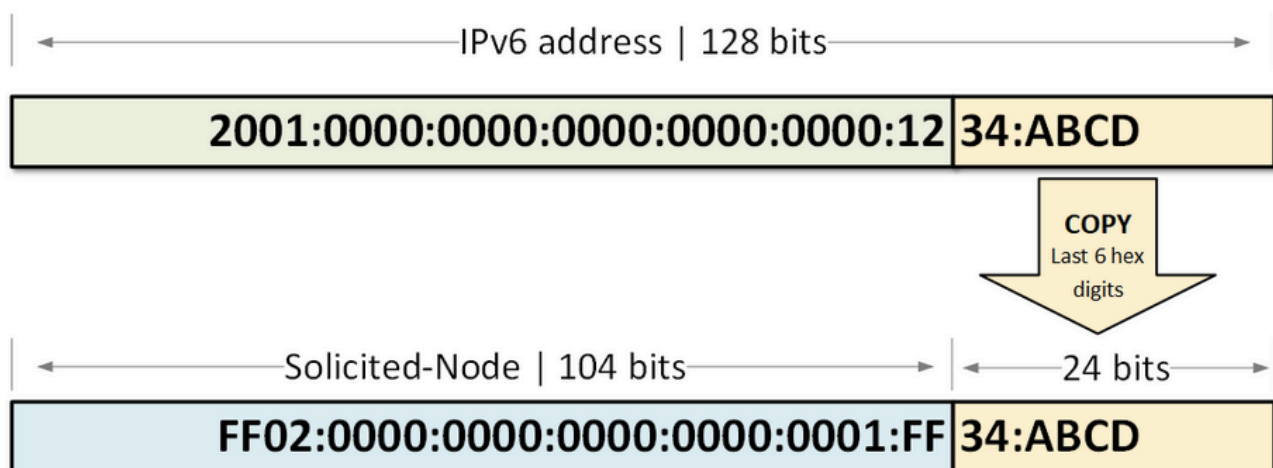
Anycast - one to nearest

- like multicast, anycast addresses identify multiple interfaces, but with a big difference
- anycast packet is delivered only one address, to the one which is based on the routing distance, the next nearest
- special address, because we can apply this to multiple interface
- one to nearest

Well-know Multicast Addresses

Address	Function
FF02::1	All Nodes Address
FF02::2	All Routers Address
FF02::4	DVMRP Routers
FF02::5	All OSPFv3 routers
FF02::6	OSPFv3 Designated Routers
FF02::a	All EIGRP (IPv6) routers
FF02::D	All PIM Routers
FF02::12	VRRP
FF02::16	All MLDv2-capable routers

Solicited-node Multicast Address



Unicast address: 2001::1234:ABCD

Multicast group: FF02::1:FF34:ABCD

- generated automatically using an IPv6 unicast of an interface
- solicited-node multicast address is generated automatically based on the unicast address for this interface and the node joins the multicast group
- any unicast address has a corresponding solicited-node multicast address
- used for address resolution, neighbor discovery, and duplicate address detection
- As shown in figure 7, a solicited-node multicast address consists of the fixed prefix FF02::1:FF00:0/104 and the last 24 bits of the corresponding IPv6 address.

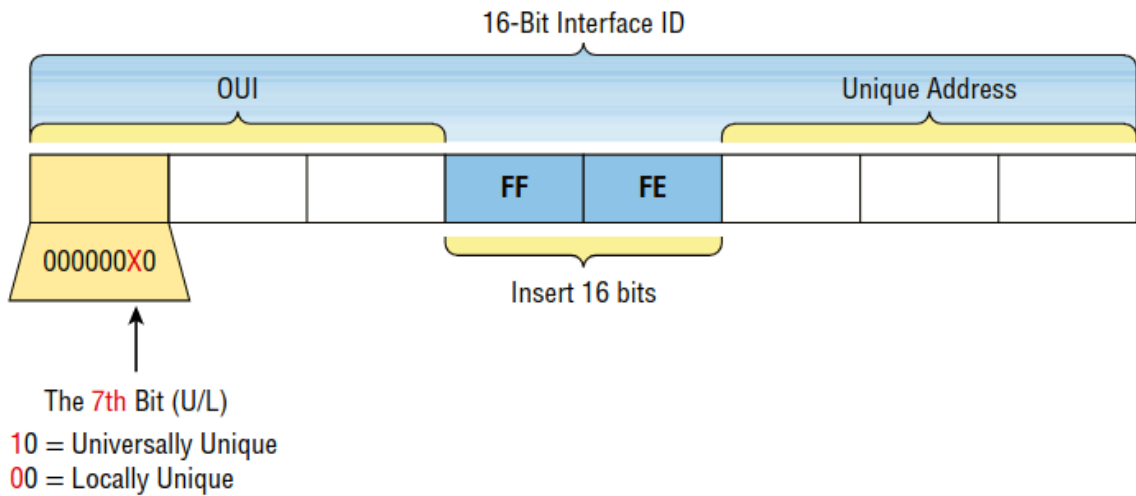
Special IPv6 Addresses

Address	Meaning
0:0:0:0:0:0:0:0	Equals <code>::</code> . This is the equivalent of IPv4's <code>0.0.0.0</code> and is typically the source address of a host before it receives an IP address (DHCP).
0:0:0:0:0:0:0:1	Equals <code>::1</code> . The equivalent of IPv4 <code>127.0.0.1</code> (loopback).
0::FFFF:192.168.100.1	IPv4-mapped IPv6 address — shows how an IPv4 address is written in a mixed IPv6/IPv4 environment.
2000::/3	The global unicast address range allocated for Internet access.
FC00::/7	The unique local unicast range (private IPv6).
FE80::/10	The link-local unicast range.
FF00::/8	The multicast range.
3FFF:FFFF::/32	Reserved for examples and documentation.
2001:0DB8::/32	Reserved for documentation (RFC 3849).
2002::/16	Used with 6to4 tunneling (IPv4-to-IPv6 automatic transition mechanism).

Stateless Address Autoconfiguration (SLAAC)

- this feature allow devices to address themselves with **local-link** or unicast address as well as with a **global unicast** address.
- happens through the first learning the prefix information form the router
- appending the device's own interfaces address as the interface ID
- MAC address used als interface ID
- but MAC is 48 bit, but interface ID is 64 bit long
- the remaining 16 bit has FF:FE value, which is placed in the middle of the MAC address
 - 0060:d673:1987 before padding
 - 0260:d6FF:FE73:1987 after padding
- so the device address is `prefix_information_from_router + device_id(padded_mac`
- fore futher information please visit or search for [#EUI-64](#) rules

FIGURE 7.3 EUI-64 interface ID assignment



- the 02 at the beginning of the address, which is determine if the address is Universal or Local (U/L) locally unique or globally unique
- its called modified EUI-64 format, using the 7th bit of the address
 - if the bit set to 1:
 - the address is generated from the MAC address
 - its become longer and less user friendly
 - its means, the address is global unique, routable
 - if the bit set to 0:
 - the address is generated by user manually, without using the MAC address
 - its become shorter, more user friendly
 - but only local valid, not routable

Original Address (hex)	Octet Before Flipping	Octet After Flipping	New Address (hex)
00:1A:2B:3C:4D:5E (MAC)	02 (binary 00000010)	03 (binary 00000011)	02:1A:2B:3C:4D:5E 03:1A:2B:3C:4D:5E
00:1A:2B:3C:4D:5E (MAC)	02	03	00:1A:2B:3C:4D:5E 00:1A:2B:3C:4D:5E U/L bit in octet 7)
Example address: 2001:0db8::/64	Interface ID: 02:00:5E:00:53:00	Flipped: 03:00:5E:00:53:00	2001:0db8::03:00

DHCP Stateful

- using it based on the SLAAC isn't necessary, but
- for take the control over the network in centralized way its possible
-

Reasons to use the DHCPv6

1. **Centralized address management:** Assign and manage IP addresses from a central location.
2. **Network segmentation (VLANs):** Assign addresses based on VLANs or segments to ensure isolation and security.
3. **Access control and security:** Enforce policies, assign specific DNS servers, gateways, or other network parameters.
4. **Ease of management:** Simplify device onboarding, reduce administrative tasks, and minimize errors.
5. **Address pool management:** Allocate and deallocate addresses from a pool to manage address availability and conservation.
6. **Client configuration:** Provide clients with specific configuration parameters, such as DNS servers, gateways, or NTP servers.
7. **Quality of Service (QoS):** Assign addresses based on QoS policies to prioritize traffic.
8. **Mobility support:** Enable devices to move between networks and obtain new addresses without manual configuration.
9. **Stateful and stateless operation:** Operate in either stateful or stateless mode, depending on the client's needs.
10. **Scalability:** Support large numbers of clients and address assignments.

These reasons highlight the key benefits of using DHCPv6 for managing IP address assignments in IPv6 networks.

Migrating IPv6 - Strategies

- [Dual Stacking](#) - its allow to use both addressing simultaneously, if they implemented
- [6to4 Tunneling](#) - used by IPv6 network, that are using IPv4 network to reach another IPv6 network

Dual Stacking

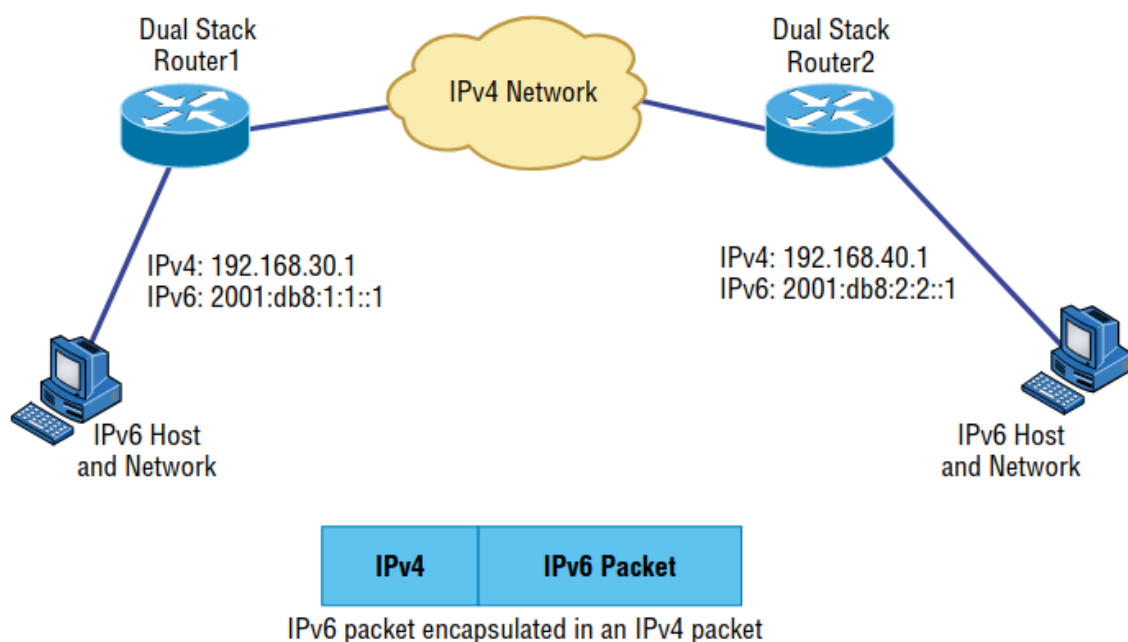
- its allow us to use both addressing to communicate

- let us upgrade the devices one by one
- during the process, that more and more device has the IPv6 address, its become more and more used over the IPv4 stack
- once all devices are upgraded, the old IPv4 can be removed

6to4 Tunneling

- crepy, but sometimes we dont have the control over the whole network, eg. the WAN using still IPv4
- at certain point the IPv6 header will be changed to IPv4

FIGURE 7.4 A 6to4 tunnel



- dual stack router needed to implement it
- one side is 6to4 on the opposite is 6to4
- if the 6to4 tunnel reach a NAT translation point, it will brake tunnel encapsulation
 - many other kind of protocol already migrated in NAT, but a lot of instance has an older version, which means trouble
- the only solution to pass this nowadays, is using Teredo, which pack our traffic in UDP packets, which can survive the NAT
- **#Miredo** is a tunneling technique, which is used on the native IPv6 Linux and BSD Unix machines, to communicate over IPv4 directly without a dual-stack router or 6to4 tunnel

